

JESS Thermodynamic Database v8.9

Chemical species in reactions with Berkelium

24-Jan-23

Charge	CAS	Count	Molecular formula	Mol. mass
Acetic-1 Acetate ion				
-1	71-50-1	558	C(2)H(3)O(2)	59.0446
Bk+2				
2	---	2	Bk(1)	247.000
Bk+3 Berkelium(III) ion				
3	22541-47-5	29	Bk(1)	247.000
Bk+3_Acetic-1				
2	---	1	C(2)H(3)Bk(1)O(2)	306.045
Bk+3_Br-1				
2	---	1	Bk(1)Br(1)	326.904
Bk+3_CDTA-4				
-1	---	1	C(14)H(18)Bk(1)N(2)O(8)	589.306
Bk+3_Citric-3				
0	---	3	C(6)H(5)Bk(1)O(7)	436.102
Bk+3_Citric-3(2)				
-3	---	2	C(12)H(10)Bk(1)O(14)	625.203
Bk+3_Cl-1				
2	---	1	Bk(1)Cl(1)	282.453
Bk+3_CO3-2				
1	---	1	C(1)Bk(1)O(3)	307.009

Bk+3_CO3-2 (2)				
-1	---	1	C (2) Bk (1) O (6)	367.018
Bk+3_DTPA-5				
-2	---	1	C (14) H (18) Bk (1) N (3) O (10)	635.311
Bk+3_EDTA-4				
-1	---	1	C (10) H (12) Bk (1) N (2) O (8)	535.214
Bk+3_F-1				
2	---	1	Bk (1) F (1)	265.998
Bk+3_Glycolic-1				
2	---	2	C (2) H (3) Bk (1) O (3)	322.044
Bk+3_Glycolic-1 (2)				
1	---	1	C (4) H (6) Bk (1) O (6)	397.088
Bk+3_H+1_Citric-3 (2)				
-2	---	2	C (12) H (11) Bk (1) O (14)	626.211
Bk+3_OH-1				
2	---	1	Bk (1) H (1) O (1)	264.007
Bk+3_Oxalic-2				
1	---	2	C (2) Bk (1) O (4)	335.020
Bk+3_Oxalic-2 (2)				
-1	---	2	C (4) Bk (1) O (8)	423.039
Bk+3_SCN-1				
2	---	2	C (1) Bk (1) N (1) S (1)	305.078
Bk+3_SO4-2				
1	---	1	Bk (1) O (4) S (1)	343.058

Bk+3_SO4-2 (2)				
-1	---	1	Bk (1) O (8) S (2)	439.115
Bk+3_SO4-2 (3)				
-3	---	1	Bk (1) O (12) S (3)	535.173
Bk+3_TriMDTA-4				
-1	---	1	C (11) H (14) Bk (1) N (2) O (8)	549.241
Bk+4 Berkelium(IV) ion				
4	22541-68-0	1	Bk (1)	247.000
Bk (s) Berkelium				
0	7440-40-6	3	Bk (1)	247.000
Br-1 Bromide ion				
-1	24959-67-9	1213	Br (1)	79.9040
C (s) Carbon; Graphite; Carbon, hexagonal				
0	7440-44-0	530	C (1)	12.0110
CDTA-4 trans-1,2-cyclohexylenedinitrilotetraacetate ion; trans-1,2-diaminocyclohexanetetraacetate ion; CDTA ion				
-4	22005-54-5	301	C (14) H (18) N (2) O (8)	342.306
Citric-3 Citrate ion				
-3	126-44-3	1347	C (6) H (5) O (7)	189.102
Cl-1 Chloride ion				
-1	16887-00-6	2242	Cl (1)	35.4530
CO3-2 Carbonate ion				

-2	3812-32-6	1435	C(1)O(3)	60.0092
DTPA-5 Diethylenetrinitrilotetraacetate ion; 1,4,7-triazaheptane-1,1,7,7-pentaacetate ion; N,N-bis[2-[bis(carboxymethyl)amino]ethyl]glycinate ion; DTPA ion				
-5	---	343	C(14)H(18)N(3)O(10)	388.311
e-1 Electron				
-1	---	3399	E(1)	0.000000
EDTA-4 Ethylenediaminetetraacetate anion				
-4	---	1457	C(10)H(12)N(2)O(8)	288.214
F-1 Fluoride ion				
-1	16984-48-8	1090	F(1)	18.9984
Glycolic-1 Glycolate ion				
-1	57122-18-6	342	C(2)H(3)O(3)	75.0440
H+1 Hydrogen ion; Proton				
1	12408-02-5	31679	H(1)	1.00794
H+1_Citric-3				
-2	---	93	C(6)H(6)O(7)	190.109
N2 (g) Nitrogen gas; Nitrogen				
0	7727-37-9	566	N(2)	28.0134
OH-1 Hydroxide ion				
-1	14280-30-9	6558	H(1)O(1)	17.0073
Oxalic-2 Oxalate ion				
-2	338-70-5	1241	C(2)O(4)	88.0196

S(s) Sulphur, alpha; Sulphur, orthorhombic; Sulphur; Sulfur, alpha; Sulfur, orthorhombic; Sulfur; Sulfur-Rhmb				
0	7704-34-9	619	S(1)	32.0600
SCN-1 Thiocyanate ion; NCS-1 equivalent				
-1	302-04-5	785	C(1)N(1)S(1)	58.0777
SO4-2 Sulfate ion; Sulphate ion				
-2	14808-79-8	1270	O(4)S(1)	96.0576
TriMDTA-4				
-4	---	107	C(11)H(14)N(2)O(8)	302.241